Statistics:

Statistics is the science of collecting, organizing and analysing data, There are two types of statistics

1) **Descriptive stats**
2) **Inferential Stats**

Descriptive stats:

It consists of organizing and summarising the data**.** They provide simple summaries of the sample and the measures. Together with simple numerical and graphical analysis.

Inferential Stats:

it consists of making some conclusions using some experiments based on collected data.

With inferential statistics, you are trying to reach conclusions that extend beyond the immediate data alone. For instance, we use inferential statistics to try to infer from the sample data what the population might think.

Descriptive stats:

It consists of organizing and summarising the data**.** They provide simple summaries of the sample and the measures. Together with simple graphical analysis**.**

With descriptive statistics**,** you are simply describing what is or what the data shows

1) **Measures of Central Tendency**
2) **Measures of Dispersion**
3) **Measures of Distribution Shape**
4) **Measures of Association**
5) **Frequency Distribution**
6) **Graphical Representation**

Measure of central tendency

a. **Mean**
b. **Median**
c. **Mode**

Each of these measures describes a different indication of the typical or **central value** in the distribution.

Mean: Find out the **Average** of the data

Find mean of [2,3,4,5,6]

$$x = \frac{2+3+4+5+6}{5}$$

$$x = \frac{20}{5}$$

$$x = 4$$

Mode: Mode is the **most frequently** occurring element.

Example1: [2,2,2,3,4,4,5,5,5]

2 and 5 is a mode for the above case

Example2: [2,2,2,3,4,4,5,5,7,8,8]

2 is the mode in this case since 2 is occurring most of the time.

Median: It will find the **central Element.**

Median for Even:

If the Number of elements are even the find the average of the central elements.

Ex [3,4,**5,6**,8,100]

Median for Even

=5+6/2

=11/2

=5.6

Median for Odd:

If the Number of elements are odd then find the average of the central element.

Ex [3,4,**5,**6,120]

We can directly take the central element as the final value for the Median which is 5 in this case.

Measure of dispersion:

The measures of central tendency are not adequate to describe data. Two data sets can have the same mean but they can be entirely different. Thus to describe data, one needs to know the extent of variability. This is given by the measures of dispersion.

1) **Variance**
2) **Standard deviation**
3) **Percentile and quartile**
4) **interquartile range (IQR)**

Variance:

Variance measures the average squared deviation of each data point from the mean. Variance talks about the spread of data. There are two types of variance,

1) Population Variance
2) Sample Variance.

Population Variance(σ^2):

When the data size is small then the population variance is calculated on the actual population dataset.
Calculate the Population variance for the below dataset.

X [1,2,3,4,5]

$$x = \frac{\sum_{I=1}^{N}(X_I - u)^2}{N}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}$$

$$x = 2$$

Sample Variance(σ^2):

When the number of observations increases then a few data points are selected that can represent the entire population. These specific data points form a sample and the variance calculated on this data is called the sample variance.
Calculate the Population variance for the below dataset.
Consider the above(X) sample data is taken from the population data

X [1,2,3,4,5]

$$x = \frac{\sum_{I=1}^{N}(X_I - u)^2}{N-1}$$

$$x = \frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{4}$$

$$x = 2.5$$

Standard Deviation(σ):

It is statistical measure of the amount of **dispersion** in a set of values. It measures how spread out the values in a dataset are from the mean (average) value. A low standard deviation indicates that the values tend to be close to the mean, while a high standard deviation indicates that the values are spread out over a wider range.

Population STD

$x = [1,2,3,4,5]$

$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{5}}$$

$x = \sqrt{2}$

$x = 1.41$

Sample STD

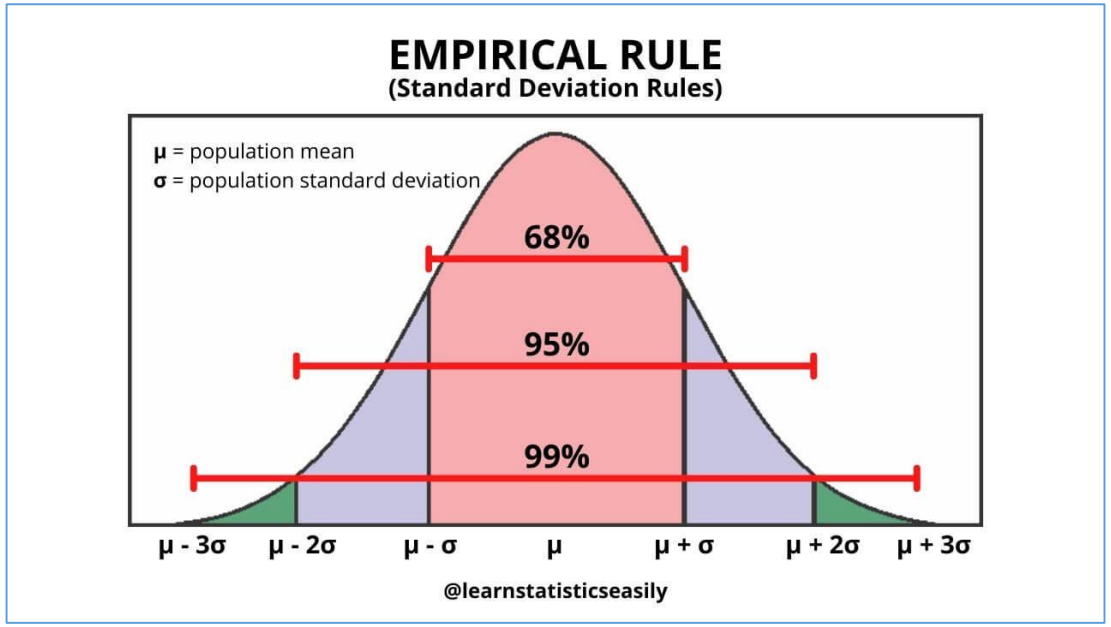
$x = [1,2,3,4,5]$

$$x = \sqrt{\frac{\sum_{I=1}^{N}(X_I - u)^2}{N-1}}$$

$$x = \sqrt{\frac{(1-3)^2+(2-3)^2+(3-3)^2+(4-3)^2+(5-3)^2}{4}}$$

$x = \sqrt{2.5}$

$x = 1.581$



Normally Disturbed Data.

Percentile:

A percentile is a statistical measure used to describe the relative position of a particular value in a dataset. It tells you what percentage of data points are below or equal to a given value.

1) Sort the value
2) Use below equation to calculate the Percentile

$$x = \frac{n}{m} * 100$$

$n = Number\ of\ Value\ till\ x$
$m = Total\ number\ of\ Value$

Calculate the percentile of value 8 on below data set

x = [5, 4, 6, 10, 1, 7, 8, 3, 9, 2]

x = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10]

$$x = \frac{n}{m} * 100$$

$$x = \frac{8}{10} * 100$$

$$x = 80\ \%$$

Calculate the percentile of below data set

x = [77,83,97,85,74,70,98,91,94,78,65,40,69,72,94,60,76,49]

x = [40,49,60,65,69,70,72,74,76,77,78,83,85,91,94,94,97,98]

$$x = \frac{n}{m} * 100$$

Percentile of number 49

$$x = \frac{2}{18} * 100$$

$$x = 11.11\%$$

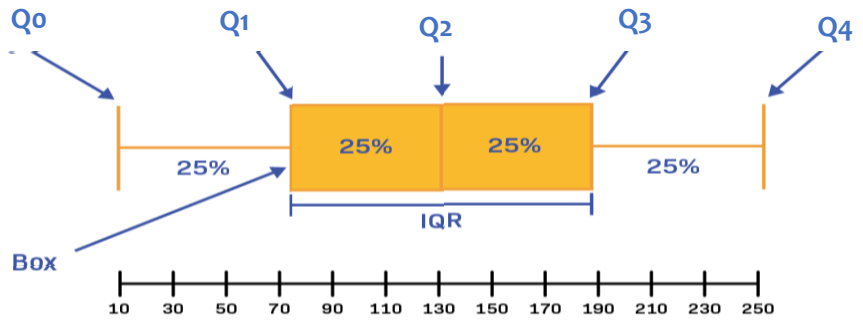
Percentile of number 77

$$x = \frac{10}{18} * 100$$

$$x = 55.55\ \%$$

Quartile:

Quartiles are a specific type of percentile that divides a dataset into four equal parts or quarters. They are often used to understand the distribution and spread of data. Quartiles are useful for identifying the central tendency and variability of a dataset, as well as for detecting potential outliers.
There are three quartiles commonly calculated:



First Quartile (Q1):

This is the 25th percentile, also known as the lower quartile. It divides the lowest 25% of the data from the rest. In other words, Q1 represents the value below which 25% of the data points fall.

Second Quartile (Q2):

This is the 50th percentile, which is also known as the median. Q2 divides the data into two equal halves, with 50% of the data below it and 50% above it. The median is often used to measure the central tendency of a dataset.

Third Quartile (Q3):

This is the 75th percentile, also known as the upper quartile. It divides the lowest 75% of the data from the highest 25%. Q3 represents the value below which 75% of the data points fall.

```
Inter Quartile Range(IQR)  = Q3 - Q1
Lower Fence                = Q1 - 1.5 * IQR
Higher Fence               = Q3 + 1.5 * IQR
```

Measures of Distribution Shape:

When analyzing data, measures of distribution shape help us understand the way the data is spread out or clustered. The most common measures of distribution shape include:

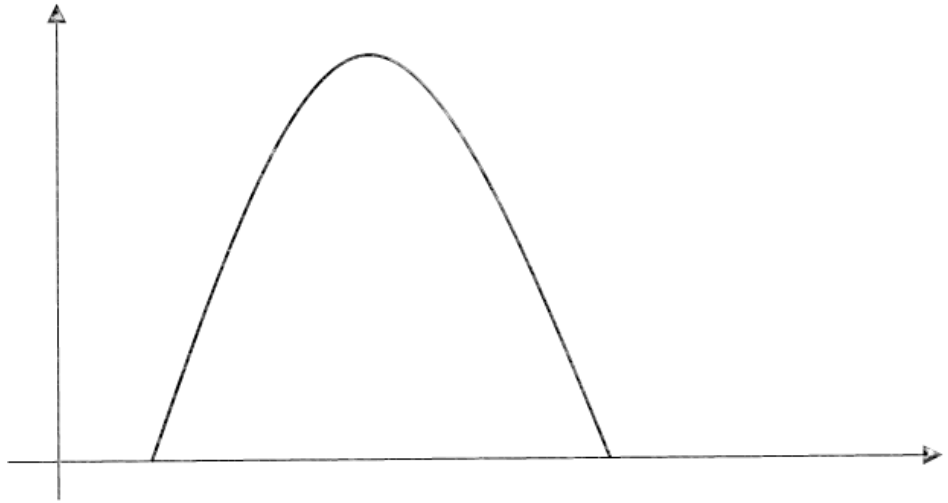
Skewness	Kurtosis
• Left • Right	• Positive • Negative

Skewness:
A measure of the asymmetry of the distribution.

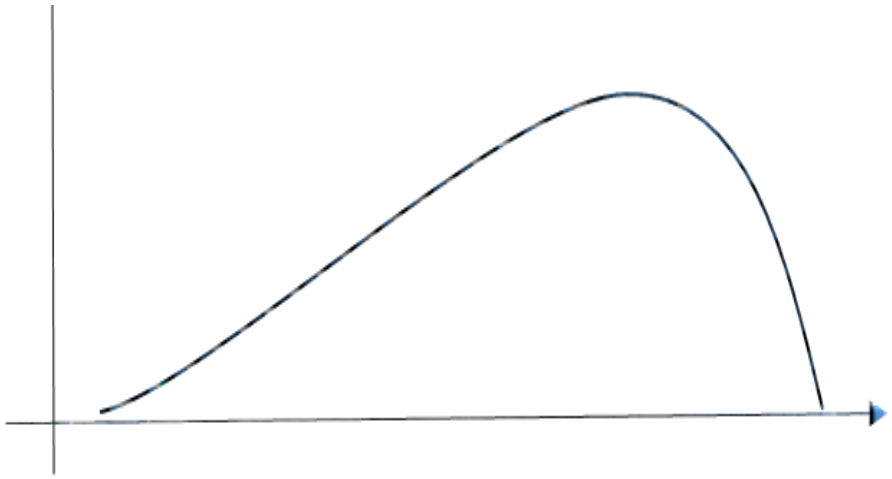
Negative skewness indicates a longer tail on the left side.

Positive skewness indicates a longer tail on the right side of the distribution,

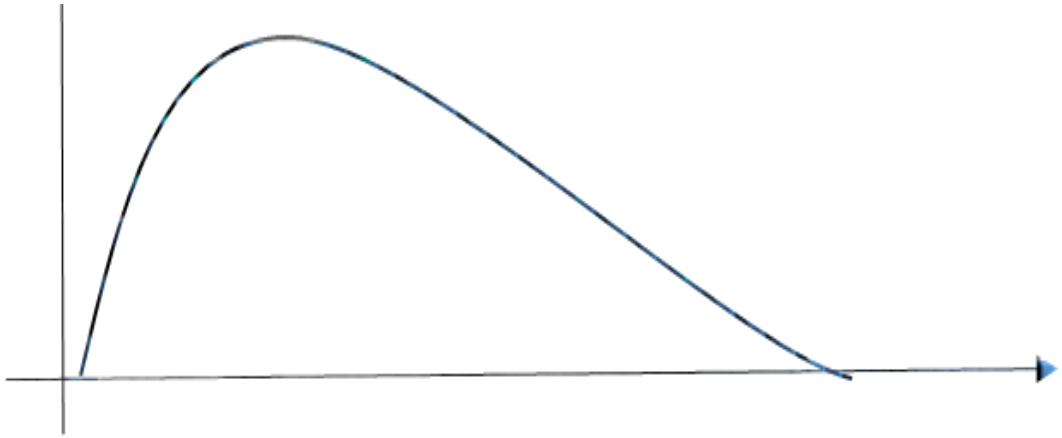
Normal Distribution, no tail on either side



Normally distributed (Gaussian Curve)



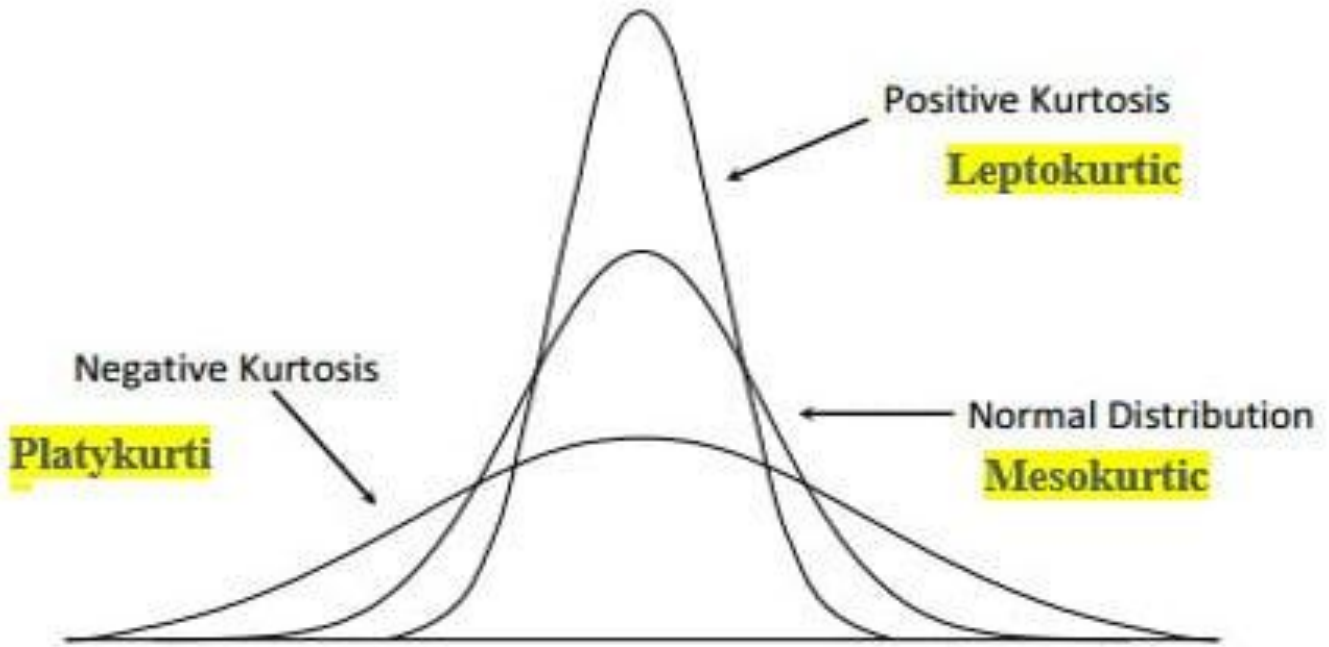
Left(Negative) Skewness



Right(Positive) Skewness

Kurtosis:

- Higher kurtosis indicates sharper, more peaked distributions,
- while lower kurtosis indicates flatter distributions.



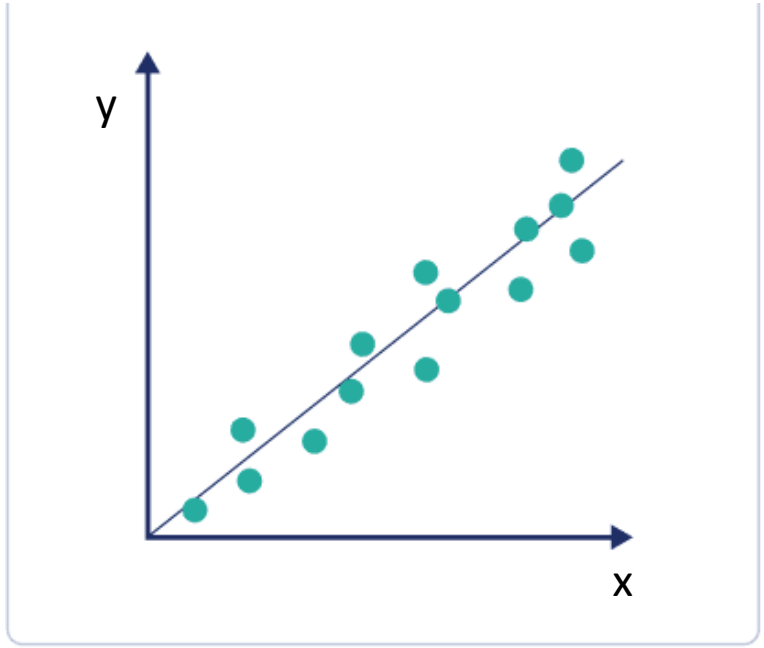
Measures of Association(Covariance) :

It is used to quantify the relationship or association between variables in a dataset. A measure of how much two variables change together. It indicates the degree to which two variables vary together.

+ Ve Covariance

X is Increasing and Y is Increasing

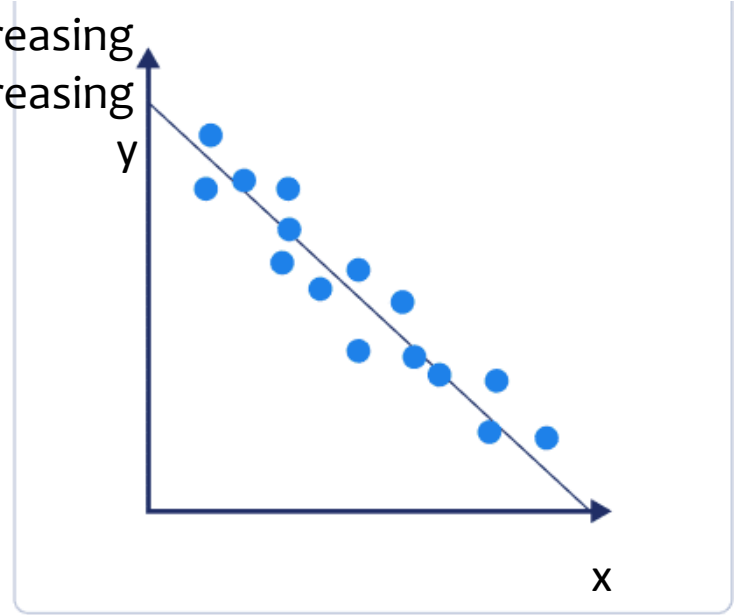
X is Decreasing and Y is Decreasing



-Ve Covariance

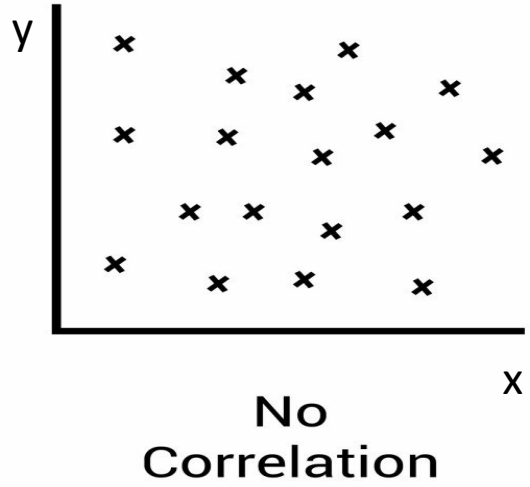
X is Increasing and Y is Decreasing

X is Decreasing and Y is Increasing



Covariance 0:

No relation between X and Y

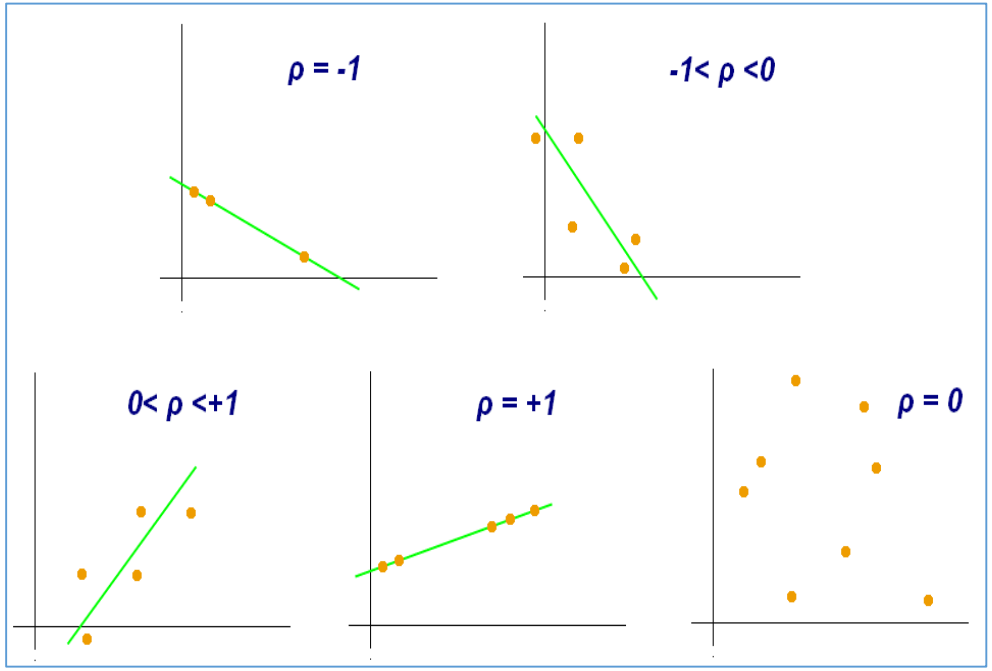


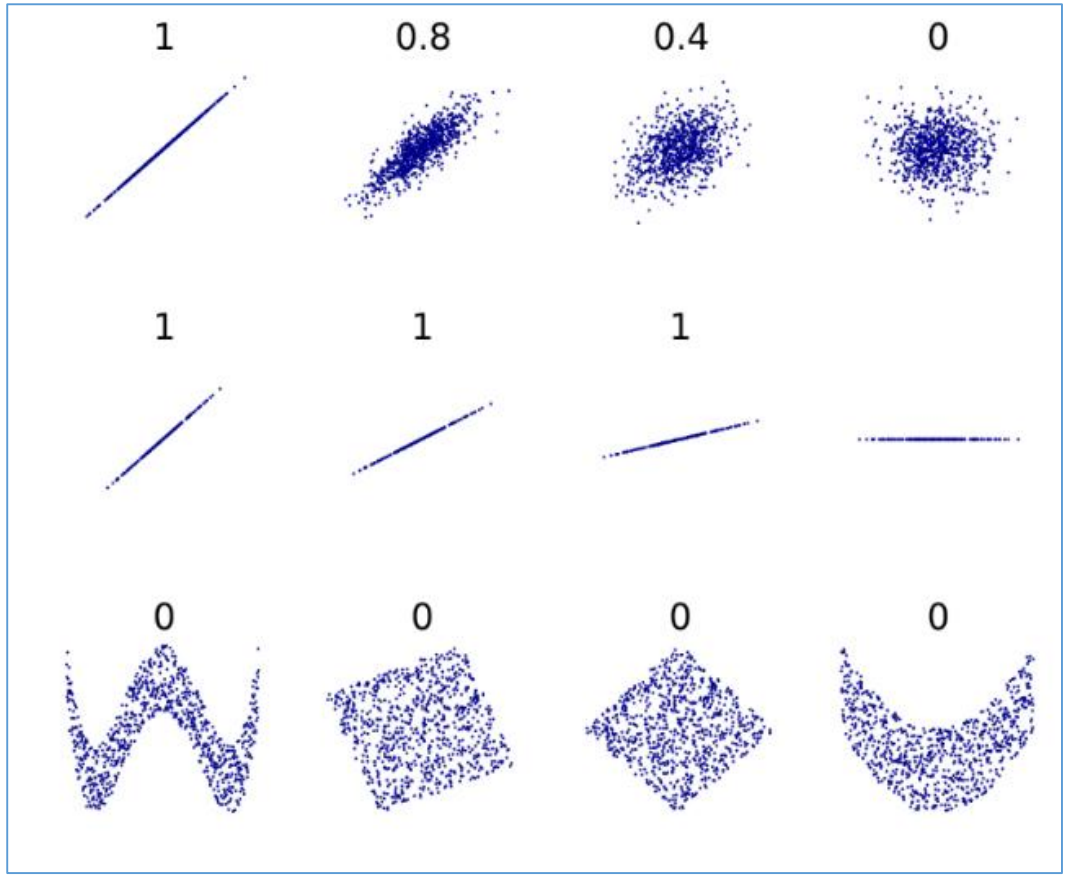
Correlation Coefficient :

Correlation estimates the depth of the relationship between variables. The correlation coefficient is a measure of the strength and direction of the linear relationship between two variables. It quantifies the degree to which changes in one variable are associated with changes in another variable.
There are several types of correlation coefficients, including
1) Pearson correlation coefficient,
2) Spearman rank correlation coefficient, and
The relationship between the correlation coefficient and covariance is given by;

Pearson correlation coefficient:

Restricting the values between(-1 to +1)
The more the values toward the +1 more positively correlated it is and vice-versa.



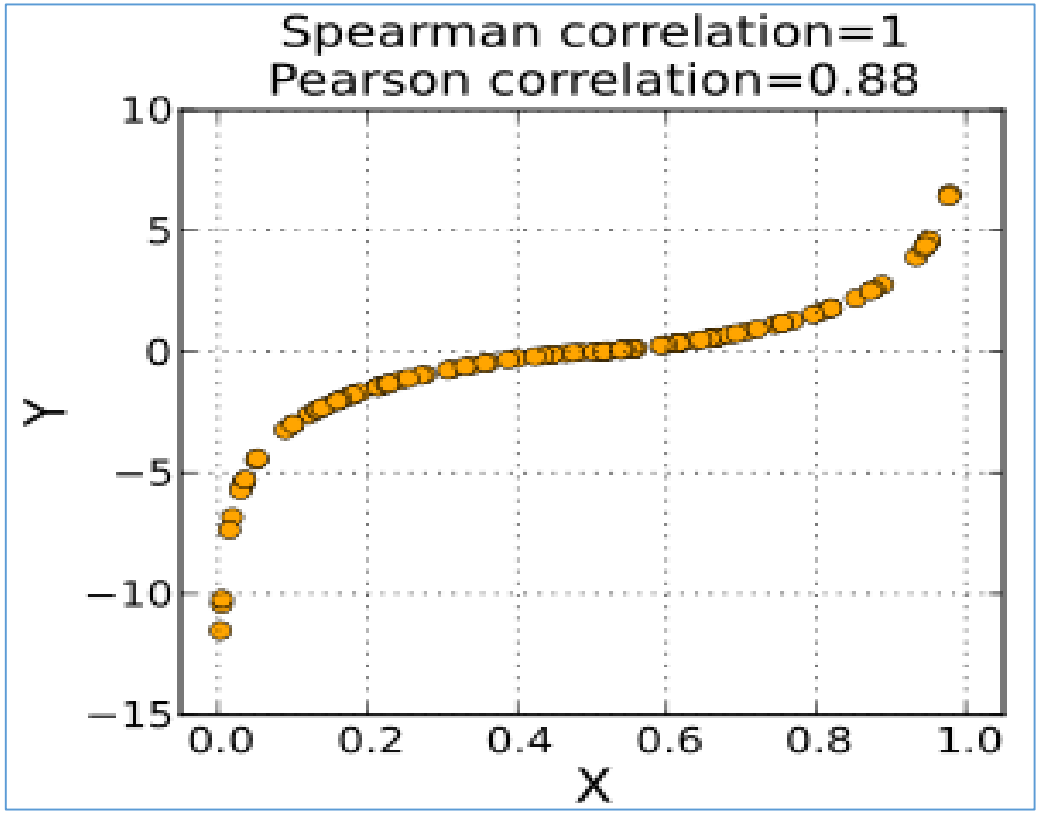


Spearman correlation:

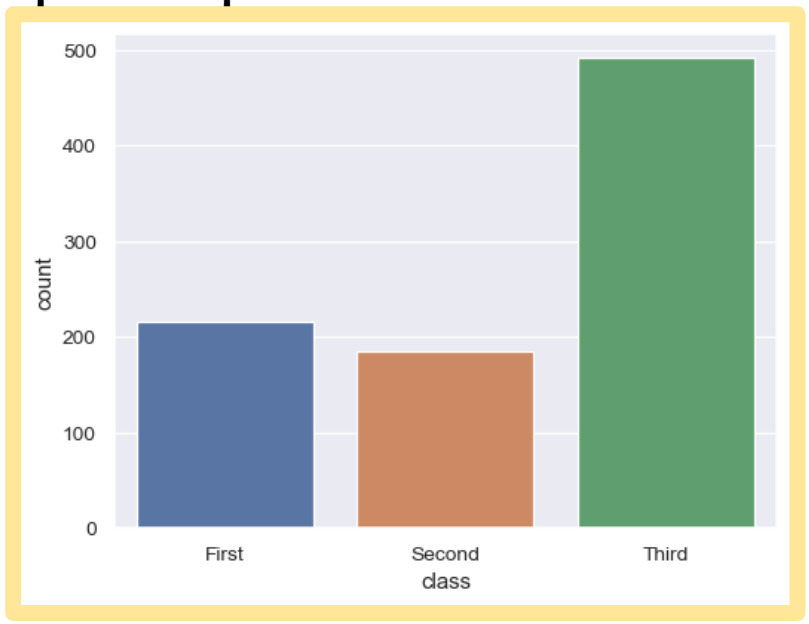
It is used only for non-linear data.

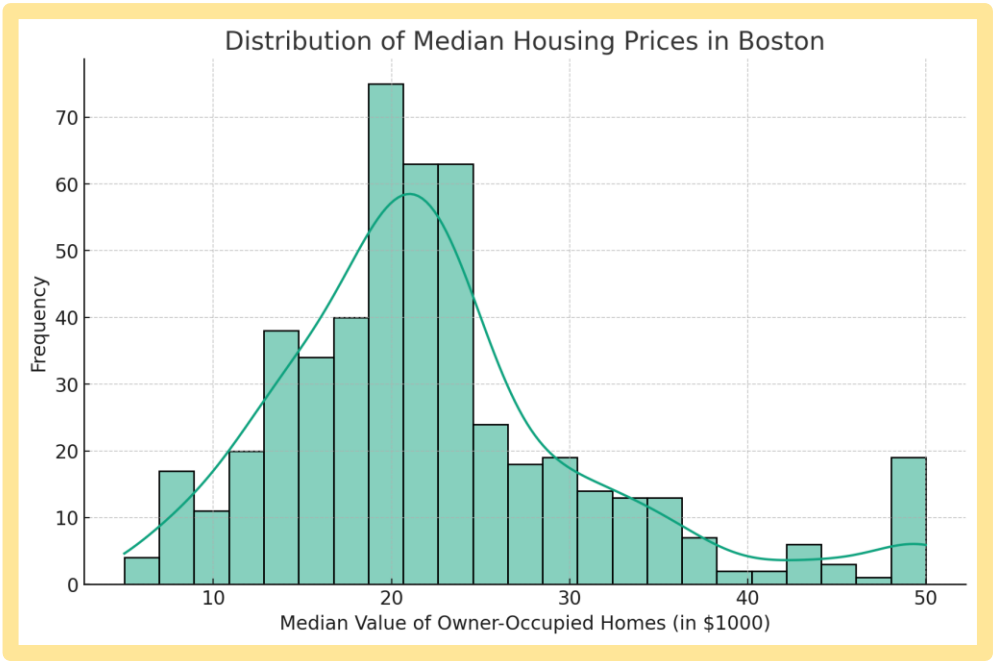
It assigns ranks

Spearman's rank correlation measures the strength and direction of association between two ranked variables. It basically gives the measure of monotonicity of the relation between two variables i.e. how well the relationship between two variables could be represented using a monotonic function.

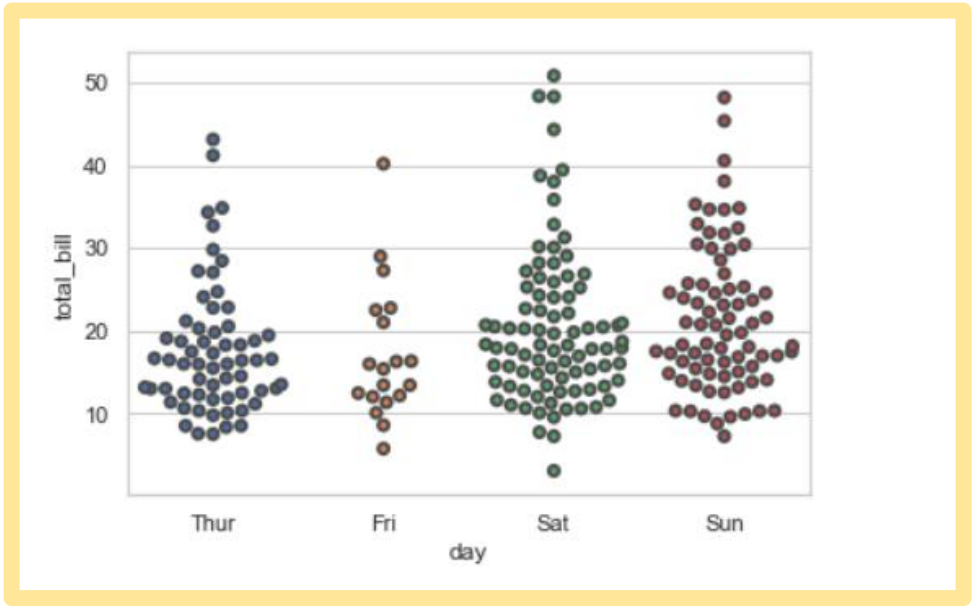


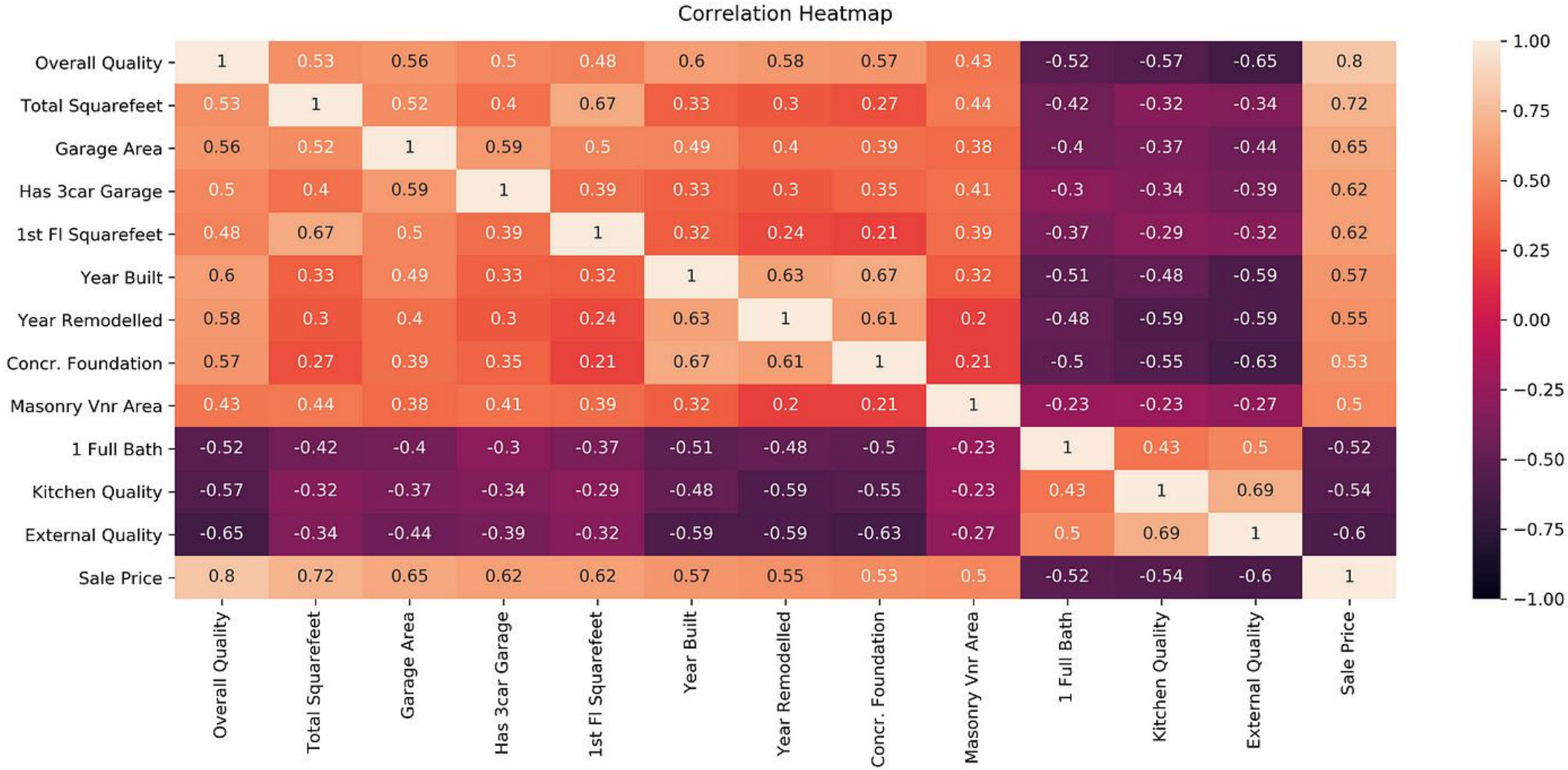
Graphical Representation

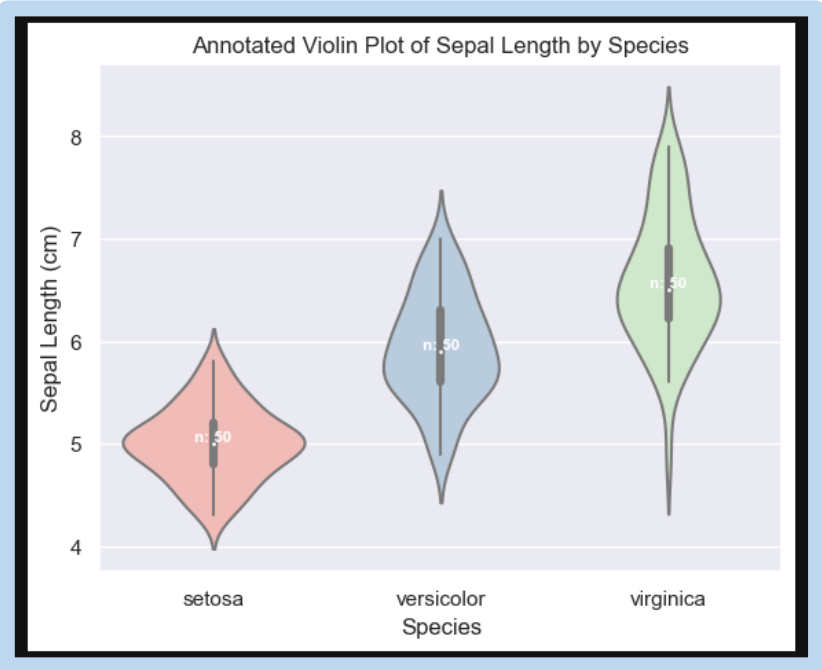


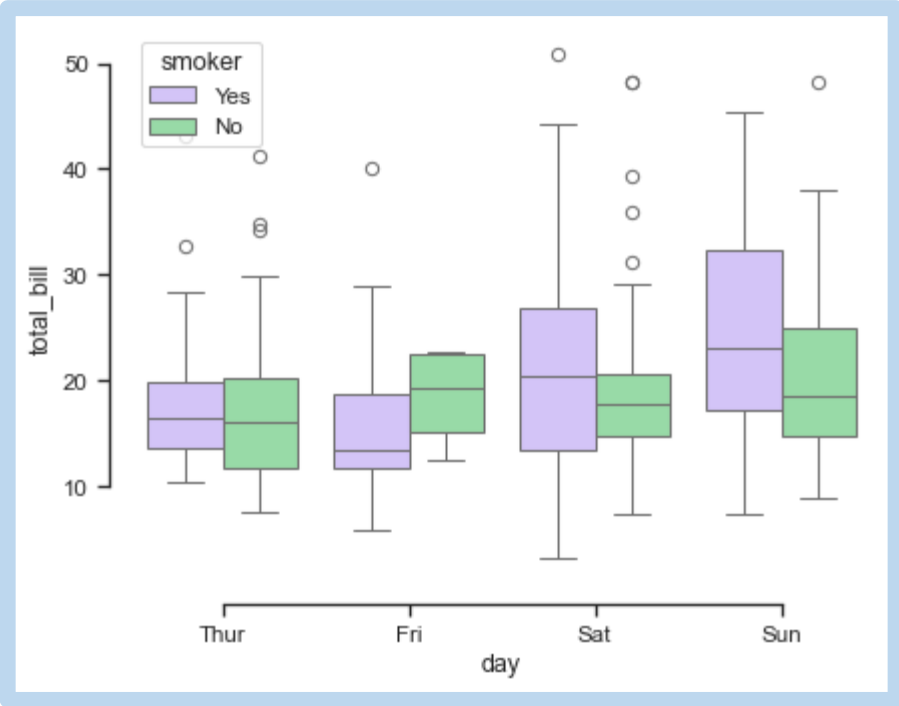


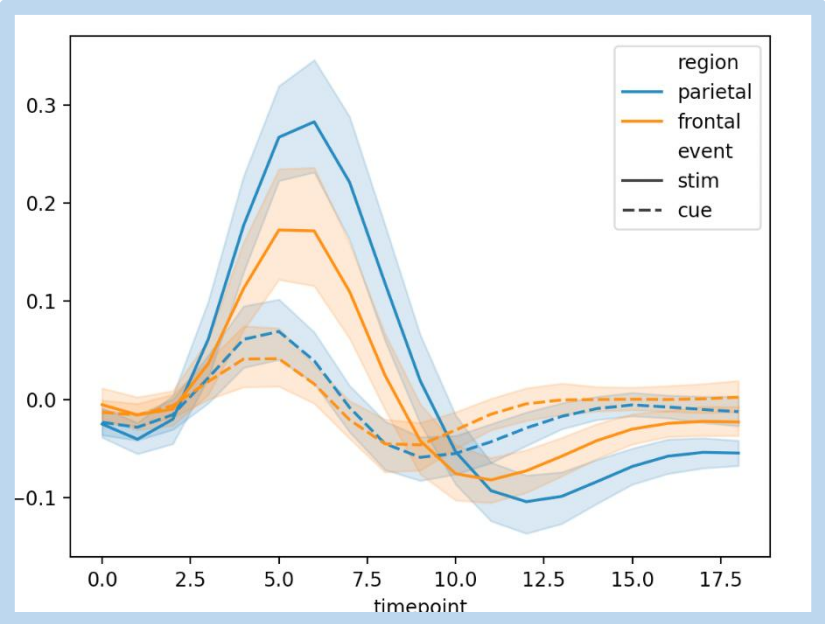


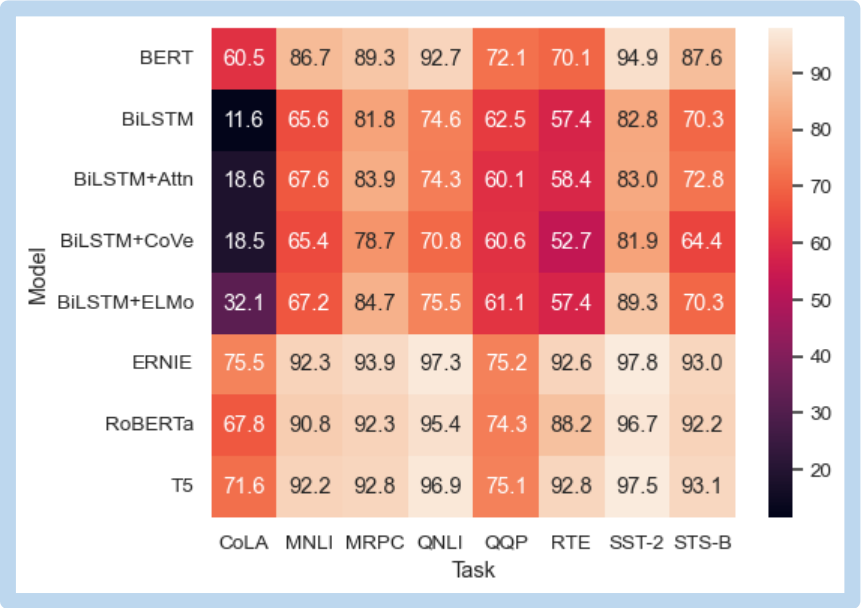



Correlation Heatmap
Overall Quality 1 0.53 0.56 0.5 0.48 0.6 0.58 0.57 0.43 -0.52 -0.57 -0.65 0.8
Total Squarefeet 0.53 1 0.52 0.4 0.67 0.33 0.3 0.27 0.44 -0.42 -0.32 -0.34 0.72
Garage Area 0.56 0.52 1 0.59 0.5 0.49 0.4 0.39 0.38 -0.4 -0.37 -0.44 0.65
Has 3car Garage 0.5 0.4 0.59 1 0.39 0.33 0.3 0.35 0.41 -0.3 -0.34 -0.39 0.62
1st Fl Squarefeet 0.48 0.67 0.5 0.39 1 0.32 0.24 0.21 0.39 -0.37 -0.29 -0.32 0.62
Year Built 0.6 0.33 0.49 0.33 0.32 1 0.63 0.67 0.32 -0.51 -0.48 -0.59 0.57
Year Remodelled 0.58 0.3 0.4 0.3 0.24 0.63 1 0.61 0.2 -0.48 -0.59 -0.59 0.55
Concr. Foundation 0.57 0.27 0.39 0.35 0.21 0.67 0.61 1 0.21 -0.5 -0.55 -0.63 0.53
Masonry Vnr Area 0.43 0.44 0.38 0.41 0.39 0.32 0.2 0.21 1 -0.23 -0.23 -0.27 0.5
1 Full Bath -0.52 -0.42 -0.4 -0.3 -0.37 -0.51 -0.48 -0.5 -0.23 1 0.43 0.5 -0.52
Kitchen Quality -0.57 -0.32 -0.37 -0.34 -0.29 -0.48 -0.59 -0.55 -0.23 0.43 1 0.69 -0.54
External Quality -0.65 -0.34 -0.44 -0.39 -0.32 -0.59 -0.59 -0.63 -0.27 0.5 0.69 1 -0.6
Sale Price 0.8 0.72 0.65 0.62 0.62 0.57 0.55 0.53 0.5 -0.52 -0.54 -0.6 1
Overall Quality
Total Squarefeet
Garage Area
Has 3car Garage
1st Fl Squarefeet
Year Built
Year Remodelled
Concr. Foundation
Masonry Vnr Area
1 Full Bath
Kitchen Quality
External Quality
Sale Price
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00


Annotated Violin Plot of Sepal Length by Species
Sepal Length (cm)
8
7
6
5
4
n: 50
n: 50
n: 50
setosa
versicolor
virginica
Species


smoker
Yes
No
total_bill
50
40
30
20
10
Thur
Fri
Sat
Sun
day


region
parietal
frontal
event
stim
cue
0.3
0.2
0.1
0.0
−0.1
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
timepoint


Model
BERT 60.5 86.7 89.3 92.7 72.1 70.1 94.9 87.6
BiLSTM 11.6 65.6 81.8 74.6 62.5 57.4 82.8 70.3
BiLSTM+Attn 18.6 67.6 83.9 74.3 60.1 58.4 83.0 72.8
BiLSTM+CoVe 18.5 65.4 78.7 70.8 60.6 52.7 81.9 64.4
BiLSTM+ELMo 32.1 67.2 84.7 75.5 61.1 57.4 89.3 70.3
ERNIE 75.5 92.3 93.9 97.3 75.2 92.6 97.8 93.0
RoBERTa 67.8 90.8 92.3 95.4 74.3 88.2 96.7 92.2
T5 71.6 92.2 92.8 96.9 75.1 92.8 97.5 93.1
CoLA MNLI MRPC QNLI QQP RTE SST-2 STS-B
Task
90
80
70
60
50
40
30
20

Estimation:

In Estimation The goal is to make educated guesses or predictions about the characteristics of a population based on information gathered from a sample.

Point Estimation:

Estimating a population parameter (e.g., mean, proportion) using a single value based on sample data.

Point estimation involves using a single value, typically derived from sample data, to estimate the value of an unknown population parameter. For example, if we want to estimate the population mean, we might use the sample mean as our point estimate. Similarly, if we want to estimate the population proportion, we might use the sample proportion.

Example: If we calculate the mean height of a sample of 100 individuals and find it to be 170 cm, we might use this value (170 cm) as our point estimate for the population's mean height.

Interval Estimation:

Estimating a range of values within which the population parameter is likely to fall, along with a level of confidence.

Interval estimation involves providing a range of values, known as a confidence interval, within which the true value of the population parameter is likely to lie. Unlike point estimation, which provides a single estimate, interval estimation acknowledges the uncertainty inherent in estimation by providing a range of plausible values.

Example: If we calculate a 95% confidence interval for the population mean height based on our sample data, we might find that the interval ranges from 165 cm to 175 cm. This means that we are 95% confident that the true population mean height falls within this interval.